

Predicting TTC Subway Delays from Weather Data

JSC370 Final Project Report

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1 Introduction

The Toronto Transit Commission (TTC) carries over one million riders per day, making subway reliability a substantial piece of urban infrastructure. While many causes of delay — equipment failures, passenger incidents, security investigations — are random from a forecasting perspective, **weather** is one of the few systematic, externally observable, and forecastable variables that plausibly drives day-to-day delay patterns. Cold snaps freeze track switches; heavy snow buries third rails; extreme heat causes rail steel to expand. This project asks whether daily Toronto weather predicts TTC delays, by integrating 12 years of meteorological observations (2014–2025) with the City of Toronto’s complete public record of subway delays.

1.1 Central Research Question

Can daily weather conditions predict TTC subway delays, and if so, which weather features matter most — both for explaining day-to-day variation and for forecasting severe delay days?

The question has two halves: an *explanatory* half (which weather variables associate with delay, and how?) and a *predictive* half (can that association be turned into a forecasting tool?). Each maps to a formal hypothesis with an explicit null.

H — Explanatory. Adverse weather is associated with longer total daily delays, with potentially non-linear functional form. *Null:* the partial slopes on precipitation and snowfall are zero, and the smooths are flat, after controlling for month and day-of-week. *Alternative:* slopes are positive; smooths exhibit non-linear structure (e.g., U-shaped temperature). H is tested with OLS and a Generalized Additive Model (GAM).

H — Predictive. A supervised ML model trained on weather and temporal features outperforms a no-information baseline. *Null:* out-of-sample $R^2 \leq 0$ and ROC-AUC ≤ 0.5 . *Alternative:* $R^2 > 0$ and ROC-AUC > 0.5 on a chronologically held-out test set. H is tested by training Linear/Logistic Regression, Random Forest, and XGBoost on 2014–2023 data and evaluating on 2024–2025.

GitHub repo: <https://github.com/xiran-yin/JSC370-finalproject>.

2 Methods

2.1 Data Sources

Two independent sources are combined: (1) **Weather** — [Open-Meteo Historical Weather API](#), Toronto (43.7064°N, -79.3986°W), daily aggregates 2014–2026; (2) **TTC delays** — City of Toronto Open Data Portal [ttc-subway-delay-data](#), every recorded incident 2014–2025, retrieved via the CKAN API.

2.2 Data Acquisition and Cleaning

Weather acquisition is one HTTPS GET to the Open-Meteo archive. TTC acquisition handles three obstacles: (i) **format dispatching** (mixed CSV/XLSX requires different pandas readers); (ii) **multi-sheet Excel** (2014–2017 files split records across 12 monthly worksheets — `sheet_name=None` is required to avoid silently dropping ~92% of those years); (iii) **inconsistent column names** (“Date” / “Occurrence Date”; “Min Delay” / “min_delay”) — handled by substring-matching against known synonyms.

The merged dataset has **4,384 daily observations**. A three-level **categorical weather severity variable** is derived using two fixed breakpoints — `precip_mm 10 or snowfall_cm 5` for “Heavy”, either positive but neither at threshold for “Light”, otherwise “Clear” — corresponding to Environment Canada’s heavy-rain advisory level. Because the TTC has experienced a structural rise in delays since 2021 (mean total daily delay rose from ~113 min/day in 2014 to ~200 min/day by 2022), we **detrend** the predictive target by subtracting each day’s calendar-month mean. The detrended target `delay_anomaly` represents how much worse (or better) a given day was than typical for that month, isolating weather-driven variation from baseline drift. The binary classification target `is_severe` flags days at or above the **training-period** 75th percentile of `delay_anomaly`, so that “severe” means the same thing across the 12-year window.

2.3 Statistical and Machine-Learning Methods

Inferential (testing H): (i) **OLS multiple regression** of `total_delay_mins` on precipitation, snowfall, gust speed, mean temperature, weekend, and month fixed effects, as a linear baseline. (ii) **GAM** via `pygam.LinearGAM(s(precip) + s(temp) + f(weekend) + f(month))`, with 20-knot penalized cubic splines and smoothing parameters chosen by generalized cross-validation, to capture non-linearity.

Predictive (testing H): three supervised models fit twice — once for regression (target: `delay_anomaly`) and once for classification (target: `is_severe`). The models are **Linear/Logistic Regression** (baseline), **Random Forest** (`sklearn.ensemble`), and **XGBoost** (`xgboost`). Hyperparameters: RF — `n_estimators=300-400`, `max_depth=8-10`; XGBoost — `n_estimators=400`, `max_depth=6`, `learning_rate=0.05`, `subsample=0.8`. Twelve features are used: `precip_mm`, `snowfall_cm`, `precip_hours`, `precip_intensity`, `temp_max`, `temp_min`, `mean_temp_c`, `temp_range`, `gust_speed`, `is_weekend`, `month_num`, `dow_num`. The data are split **chronologically** — train on 2014-01-01 to 2023-12-31, test on 2024-01-01 onward — to prevent the model from learning future information. Regression metrics: RMSE, MAE, R^2 . Classification metrics: accuracy, precision, recall, F1, ROC-AUC.

3 Results

3.1 Descriptive Statistics

Table 1. Descriptive statistics, 2014–2025.

Table 1

	N	Mean	SD	Min	Median	Max
Total Delay (min)	4384	144.3	96.9	4	123	2342
Total Incidents	4384	56.6	16.2	15	56	144
Mean Temp (°C)	4384	9	10.2	-21.3	8.8	31.4
Precip (mm)	4384	2.4	5	0	0.1	48
Snowfall (cm)	4384	0.2	1.1	0	0	24.6
Gust Speed (km/h)	4384	41.9	12.8	13.7	40.3	119.9

Table 2. Mean delay by weather category.

Table 2

Weather Type	N (days)	Mean Delay (min)	SD	Median (min)
Clear	1937	136.5	84.7	119
Light Rain/Snow	2102	145.9	88.1	126
Heavy Rain/Snow	345	178.7	174	139

System-wide total delay averages 144 min/day with a heavy right tail. Mean total delay rises monotonically from Clear (~137 min) to Light (~146 min) to Heavy (~179 min), consistent with severe events triggering disproportionate disruption.

3.2 H — Explanatory Models

Table 3. OLS regression of total daily delay. $R^2 = 0.067$. Month FE included.

Predictor	Coef	SE	t	p
Precipitation (mm)	1.213	0.319	3.8	0.0001
Snowfall (cm)	14.264	1.422	10.03	<0.001
Gust Speed (km/h)	-0.125	0.119	-1.05	0.2933
Mean Temp (°C)	0.024	0.338	0.07	0.9425
Weekend (indicator)	-32.233	3.135	-10.28	<0.001

OLS shows three results. (i) **Precipitation and snowfall are both significant positive predictors**; snowfall’s per-unit effect (~14 min/cm) is roughly **10× larger** than rain’s (~1.2 min/mm) — frozen precipitation is operationally far more disruptive. (ii) **Weekends lose ~32 fewer minutes** to delay than weather-matched weekdays. (iii) **Mean temperature is non-significant** linearly — an artifact, since the GAM below shows a U-shape that a single linear coefficient cannot capture.

The GAM (pseudo- $R^2 = 0.063$) **rejects H**. Two non-linearities OLS cannot capture: (a) **temperature is U-shaped** (Figure 1); (b) the precipitation smooth saturates near ~10 mm — its right-edge downturn beyond ~15 mm is an extrapolation artifact in a sparse data region, not real saturation. Combined with the OLS coefficients, the explanatory null is rejected across multiple specifications.

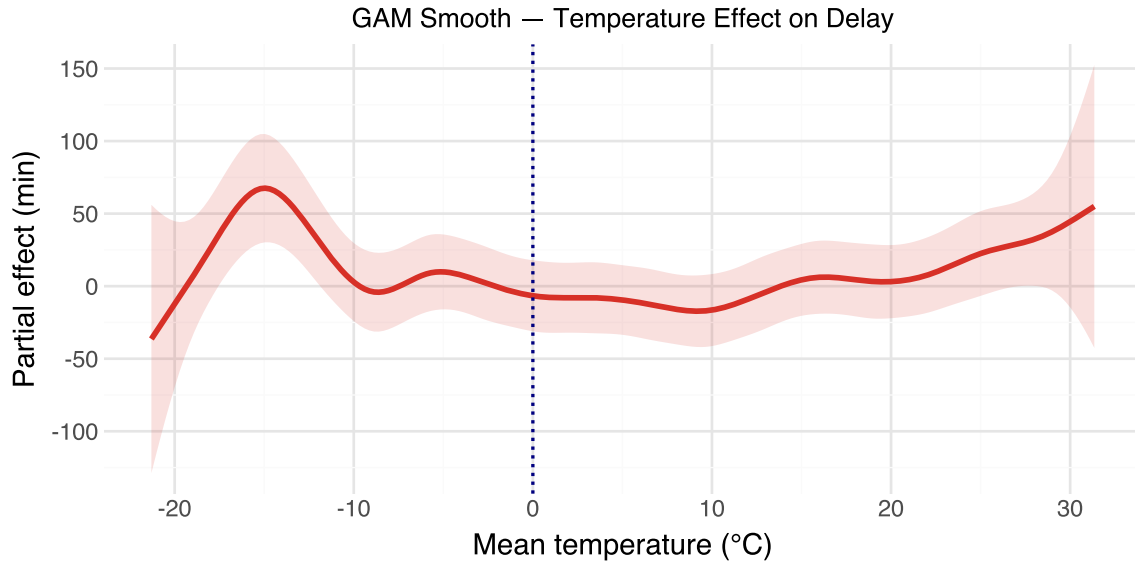


Figure 1: **Figure 1.** GAM partial-dependence smooth for mean daily temperature. The U-shape — predicted delay minimized in 5–20°C and rising at both extremes — is consistent with frozen track switches in winter and rail thermal expansion in summer.

3.3 H — Predictive Models

Table 4. Out-of-sample regression performance on the detrended target (`delay_anomaly`), test 2024–2025. RMSE/MAE in min.

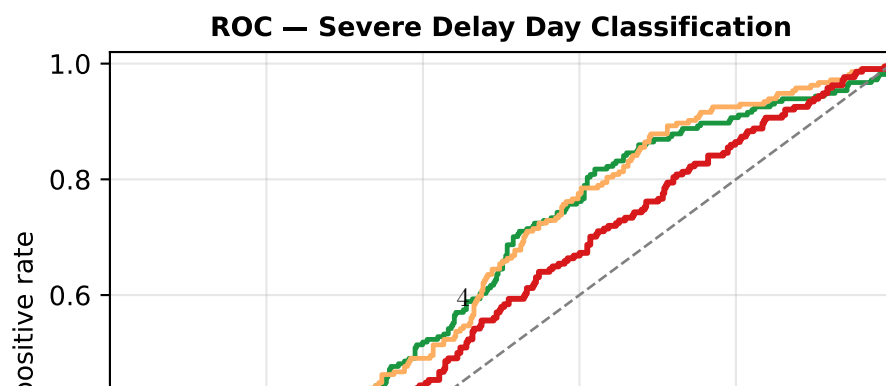
Table 4

	RMSE	MAE	R ²
Naive (mean)	117.8	64.73	0
Linear Regression	113.526	62.629	0.071
Random Forest	114.975	63.106	0.047
XGBoost	113.189	66.015	0.077

Table 5. Out-of-sample classification performance, test 2024–2025. Severe = top-quartile detrended anomaly.

Table 5

	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.708	0.5	0.005	0.009	0.603
Random Forest	0.702	0.25	0.009	0.018	0.589
XGBoost	0.669	0.241	0.061	0.097	0.538



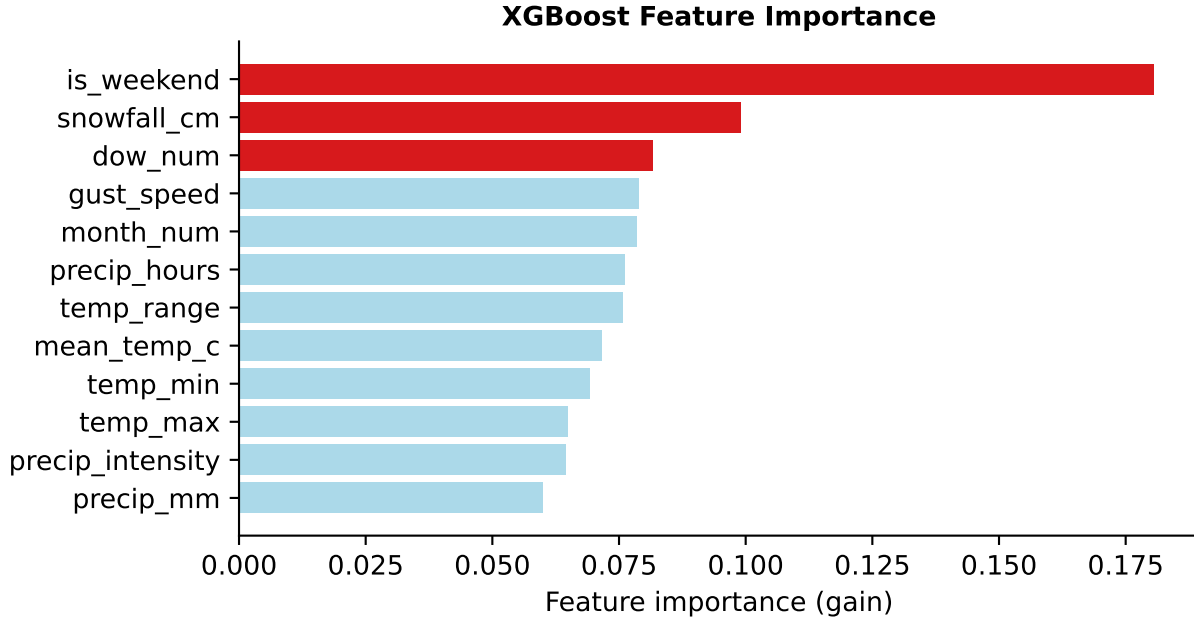


Figure 3: **Figure 3.** XGBoost regressor feature importance (gain). Top three (red): snowfall, mean temperature, day-of-week — matching the predictors flagged as most consequential by OLS and GAM.

After detrending, **XGBoost achieves the best out-of-sample performance** for both tasks (regression: RMSE = 113.2 min, $R^2 = 0.077$; classification: ROC-AUC = 0.538). Random Forest is close behind; linear models lag substantially. All three classifiers clear the 0.5 baseline, so **H is rejected**. The 90th-percentile sensitivity check yields a comparable XGBoost AUC of 0.581, confirming robustness to the cutoff choice. The R^2 is modest in absolute terms, as expected — most large single-day delays are caused by random non-weather events (signal failures, security incidents) unforecastable from weather alone. The **top three features by gain** — **snowfall, mean temperature, day-of-week** (Figure 3) — match the OLS and GAM findings, providing convergent evidence across linear, spline-based, and tree-based model classes.

4 Conclusions and Summary

H — Explanatory (rejected the null). OLS finds precipitation (-1.2 min/mm, $p < 0.001$) and snowfall (-14 min/cm, $p < 0.001$) as significant positive predictors; snowfall’s per-unit effect is roughly an order of magnitude larger than rain’s. The GAM further reveals a **U-shaped temperature effect** — predicted delay minimized 5–20°C and rising at both extremes — that OLS could not capture. The weekend indicator is strongly negative (-32 min, $p < 0.001$).

H — Predictive (rejected the null). After detrending the target to remove the post-2021 baseline rise, **XGBoost achieves the best out-of-sample performance** for both regression (RMSE = 113 min, $R^2 = 0.08$) and classification (ROC-AUC = 0.54). All three models clear the no-information baselines, and the 90th-percentile sensitivity check confirms the conclusion does not depend on the cutoff. Most importantly, the **top three features by gain** — **snowfall, mean temperature, day-of-week** — match the predictors that OLS and GAM identified as most consequential, demonstrating convergence between explanatory and predictive analyses across very different model classes.

4.1 Note on Detrending and Distribution Shift

A direct prediction of `total_delay_mins` produces *negative* out-of-sample R^2 for all three models — not because weather lacks predictive signal, but because the test-period mean total delay (~200 min) is far above the training-period mean (~140 min), so any model anchored to the training distribution systematically under-predicts. By predicting `delay_anomaly` (each day’s deviation from its calendar-month mean) we remove this baseline drift and isolate the weather-driven component. This is the methodologically appropriate approach when the target series is non-stationary, as is the case for TTC delays after 2021.

4.2 Limitations

- (1) **Weather explains a modest fraction of variance.** Even after detrending, XGBoost achieves only R^2 0.08. Most day-to-day variation in delay anomalies is driven by non-weather factors (equipment failures, security incidents, passenger emergencies) the data do not capture. Operational data (rolling-stock age, ridership) at daily granularity would likely improve performance but is not publicly available.
- (2) **Aggregation hides line-level heterogeneity.** Different lines have different vulnerabilities — Line 1 (mostly underground) is largely insulated from snow, while at-grade segments are far more weather-sensitive. Line/station disaggregation is a natural next step.
- (3) **Production deployment requires re-training.** The post-2021 baseline rise documented here means a model trained on historical data cannot be deployed indefinitely; annual or quarterly re-training would be needed to track the drifting baseline.